

2.3.1 & 2.3.2 Worksheet

Problems #1-8: Find the following limits. If the limit does not exist provide proper justification for stating that it does not.

1. $\lim_{x \rightarrow 1.5^-} \frac{e^x}{2x-3}$

$$\lim_{x \rightarrow 1.5^-} \frac{e^x}{2x-3} \rightarrow \frac{(e^{1.5})^-}{0^-} \rightarrow -\infty$$

$$\text{Therefore } \lim_{x \rightarrow 1.5^-} \frac{e^x}{2x-3} = -\infty$$

2. $\lim_{x \rightarrow 2} \frac{\sqrt{4x+1}-3}{x-2}$

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\sqrt{4x+1}-3}{x-2} &= \lim_{x \rightarrow 2} \frac{\sqrt{4x+1}-3}{x-2} \cdot \frac{\sqrt{4x+1}+3}{\sqrt{4x+1}+3} = \lim_{x \rightarrow 2} \frac{4x+1-9}{(x-2)(\sqrt{4x+1}+3)} = \lim_{x \rightarrow 2} \frac{4(x-2)}{(x-2)(\sqrt{4x+1}+3)} \\ &= \lim_{x \rightarrow 2} \frac{4}{\sqrt{4x+1}+3} = \frac{4}{\sqrt{4(2)+1}+3} = \frac{4}{6} = \frac{2}{3} \end{aligned}$$

3. $\lim_{x \rightarrow 0} \frac{\frac{1}{2} - \frac{1}{x+2}}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\frac{1}{2} - \frac{1}{x+2}}{x} &= \lim_{x \rightarrow 0} \frac{\frac{(x+2)}{2(x+2)} - \frac{2}{2(x+2)}}{x} = \lim_{x \rightarrow 0} \frac{\frac{x+2-2}{2(x+2)}}{x} = \lim_{x \rightarrow 0} \frac{x}{2x(x+2)} = \lim_{x \rightarrow 0} \frac{1}{2(x+2)} = \frac{1}{2(0+2)} = \frac{1}{4} \end{aligned}$$

$$4. \lim_{x \rightarrow -\infty} \frac{5+2x^3}{3x^3+2x-4}$$

$$\lim_{x \rightarrow -\infty} \frac{5+2x^3}{3x^3+2x-4} = \lim_{x \rightarrow -\infty} \frac{\frac{5}{x^3} + \frac{2x^3}{x^3}}{\frac{3x^3}{x^3} + \frac{2x}{x^3} - \frac{4}{x^3}} = \lim_{x \rightarrow -\infty} \frac{\frac{5}{x^3} + 2}{3 + \frac{2}{x^2} - \frac{4}{x^3}} = \frac{0+2}{3+0-0} = \frac{2}{3}$$

$$5. \lim_{x \rightarrow \infty} \frac{2-5e^{2x}}{3e^{2x}}$$

$$\lim_{x \rightarrow \infty} \frac{2-5e^{2x}}{3e^{2x}} = \lim_{x \rightarrow \infty} \frac{2}{3e^{2x}} - \frac{5e^{2x}}{3e^{2x}} = \lim_{x \rightarrow \infty} \frac{2}{3e^{2x}} - \frac{5}{3} = 0 - \frac{5}{3} = -\frac{5}{3}$$

$$6. \lim_{x \rightarrow \infty} \sqrt{x^2+x} - x$$

$$\lim_{x \rightarrow \infty} \sqrt{x^2+x} - x \cdot \frac{\sqrt{x^2+x} + x}{\sqrt{x^2+x} + x} = \lim_{x \rightarrow \infty} \frac{x^2+x-x^2}{\sqrt{x^2+x} + x} = \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2+x} + x} = \lim_{x \rightarrow \infty} \frac{\frac{x}{x}}{\frac{\sqrt{x^2+x}}{x} + \frac{x}{x}} = \lim_{x \rightarrow \infty} \frac{1}{\frac{\sqrt{x^2+x}}{x} + 1}$$

Since $x \rightarrow \infty$ we can assume that $x > 0$. Then we can say $x = \sqrt{x^2}$.

$$\lim_{x \rightarrow \infty} \frac{1}{\frac{\sqrt{x^2+x}}{x} + 1} = \lim_{x \rightarrow \infty} \frac{1}{\frac{\sqrt{x^2+x}}{\sqrt{x^2}} + 1} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{\frac{x^2+x}{x^2}} + 1} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{x}} + 1} = \frac{1}{\sqrt{1+0} + 1} = \frac{1}{2}$$

7. $\lim_{x \rightarrow -\infty} \frac{x^3 + 3x + 2}{4 - 5x^2 + 4x}$

$$\lim_{x \rightarrow -\infty} \frac{x^3 + 3x + 2}{4 - 5x^2 + 4x} = \lim_{x \rightarrow -\infty} \frac{\frac{x^3}{x^2} + \frac{3x}{x^2} + \frac{2}{x^2}}{\frac{4}{x^2} - \frac{5x^2}{x^2} + \frac{4x}{x^2}} = \lim_{x \rightarrow -\infty} \frac{x + \frac{3}{x} + \frac{2}{x^2}}{\frac{4}{x^2} - 5 + \frac{4}{x}} = \frac{-\infty}{-5} = \infty$$

8. $\lim_{x \rightarrow \infty} \frac{3 + 2 \ln(x)}{5 - 6 \ln(x)}$

$$\lim_{x \rightarrow \infty} \frac{3 + 2 \ln(x)}{5 - 6 \ln(x)} = \lim_{x \rightarrow \infty} \frac{3 + 2 \ln(x)}{5 - 6 \ln(x)} \cdot \frac{\frac{1}{\ln(x)}}{\frac{1}{\ln(x)}} = \lim_{x \rightarrow \infty} \frac{\frac{3}{\ln(x)} + 2}{\frac{5}{\ln(x)} - 6} = \frac{0 + 2}{0 - 6} = -\frac{1}{3}$$

9. Find all horizontal asymptotes on the graph of $f(x) = \frac{\sqrt{3x^4 + x}}{x^2 - 8}$.

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8} = \lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{3x^4 + x}}{x^2}}{\frac{x^2}{x^2} - \frac{8}{x^2}} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{3x^4 + x}}{x^2}}{1 - \frac{8}{x^2}}. \text{ Note that } x^2 = \sqrt{x^4} \text{ for all } x \text{ and}$$

$$\text{therefore } \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{3x^4 + x}}{x^2}}{1 - \frac{8}{x^2}} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{3x^4 + x}}{\sqrt{x^4}}}{1 - \frac{8}{x^2}} = \lim_{x \rightarrow -\infty} \frac{\sqrt{\frac{3x^4 + x}{x^4}}}{1 - \frac{8}{x^2}} = \lim_{x \rightarrow -\infty} \frac{\sqrt{3 + \frac{1}{x^3}}}{1 - \frac{8}{x^2}} = \frac{\sqrt{3 + 0}}{1 - 0} = \sqrt{3}.$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8} = \lim_{x \rightarrow \infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{3x^4 + x}}{x^2}}{\frac{x^2}{x^2} - \frac{8}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{3x^4 + x}}{x^2}}{1 - \frac{8}{x^2}}. \text{ Note that } x^2 = \sqrt{x^4} \text{ for all } x \text{ and}$$

$$\text{therefore } \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{3x^4 + x}}{x^2}}{1 - \frac{8}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{3x^4 + x}}{\sqrt{x^4}}}{1 - \frac{8}{x^2}} = \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{3x^4 + x}{x^4}}}{1 - \frac{8}{x^2}} = \lim_{x \rightarrow \infty} \frac{\sqrt{3 + \frac{1}{x^3}}}{1 - \frac{8}{x^2}} = \frac{\sqrt{3 + 0}}{1 - 0} = \sqrt{3}.$$

Therefore we can conclude that $f(x)$ has a horizontal asymptote of $y = \sqrt{3}$.